KENYA MEDICAL TRAINING COLLEGE
FACULTY OF REHABILITATIVE HEALTH SCIENCES
DEPARTMENT OF MEDICAL ENGINEERING

FINAL QUALIFYING EXAMINATION
FOR
DIPLOMA IN MEDICAL ENGINEERING TECHNOLOGY

PAPER: CLINICAL ENGINEERING II

DATE: 29th September, 2022	**TIME: 3 HOURS (9:00 A.M – 12 NOON)**

INSTRUCTIONS

1. This paper consists of:
   - Section 1 (40 Multiple Choice Questions)
   - Section 2 (8 Short Answer Questions)
   - Section 3 (1 Long Answer Question)
2. Attempt **ALL** Questions
3. Write the **EXAMINATION NUMBER** given on all the answer sheets provided and on the question paper.
4. Ensure that all examination answer scripts are handed in at the end of the examination
5. Ensure you sign the examination register provided

EXAMINATION NUMBER .....

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SECTION 1: MULTIPLE CHOICE QUESTIONS (40 MARKS)

1. What is the magnification of a direct ophthalmoscope?
   a) 10
   b) 20
   c) 15
   d) 17
2. The output from a laser is monochromatic, this means that it is
   a) Polarized
   b) Single frequency
   c) Narrow beam
   d) Multiple frequency
3. Anaesthesia can be induced by the following technique except?
   a) Intravenous injection
   b) Local anaesthetic
   c) Inhalation
   d) Vapourization
4. Which one of the following is not an operational feature of an electro-surgical?
   a) Internal safety precautions
   b) Maximum power output
   c) Working modes
   d) Clinical application
5. Which of these is not a feature of image produced by indirect ophthalmoscope
   a) Upright
   b) Unreversed
   c) Indirect
   d) Virtual
6. Which process gives the laser its special properties as an optical source?
   a) Dispersion
   b) Stimulated absorption
   c) Spontaneous emission
   d) Stimulated emission
7. The following are requirements of operating theatre light, which one is not?
   a) Adjustability
   b) Brightness
   c) Dimmable flicker
   d) Control

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8. Which one of the following errors is not applicable in ECG machine?
   a) Conduction disorders
   b) Electronic pacemakers
   c) Lead misplacement
   d) Knurling
9. A direct ophthalmoscope inspects the eye
   a) Aqueous humour
   b) Retina
   c) Fundus
   d) Fovea
10. During Laser production, what is the need to achieve population inversion?
   a) To excite most of the atoms
   b) To bring most of the atoms to ground state
   c) To achieve stable condition
   d) To reduce the time of production of laser
11. Which one of the following parameters is not applicable to a ventilator machine?
   a) Respiratory rate
   b) Tidal volume
   c) Inspiratory / expiratory ration
   d) Power failure
12. The following are types of infusion pumps except?
   a) Syringe
   b) Large volume
   c) Elastometric
   d) Tidal pump
13. Which one of the following statements is not correct about factors that determine the rate of contamination or infection in operation theatre?
   a) Size of operation theatre
   b) Sepsis techniques
   c) Drapping procedures
   d) Operation theatre skills
14. Which one of the following equipment is odd one out?
   a) Theatre table
   b) Suction machine
   c) Centrifuge
   d) Electrosurgical unit

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15. A single and fundamental form of load cell which is basically used as a transducer in the operation of a weighing machine undergoes the conversion of
    a) Force into an electric signal
    b) Pressure into an electrical signal
    c) Acceleration into an electrical signal
    d) Velocity into an electrical signal
16. While using servo mode the heater output of a radiant warmer is controlled by
    a) Machine
    b) Patient
    c) Both machine and patient
    d) None
17. What is the frequency operation range in ultrasonic therapy unit?
    a) 80KHZ to 10 MHZ
    b) 1KHZ to 12.5 MHZ
    c) 800KHZ to 1 MHZ
    d) 100KHZ to 1 MHZ
18. The purpose of relief valve in a suction pump is to
    a) Control the rate of discharge
    b) Reduce discharge pressure
    c) Protect the pump against developing excessive pressure
    d) Facilitate unidirectional flow of fluid
19. What should be used to disinfect radiant warmer after each use?
    a) 2.5% bacillocid
    b) Soap and water
    c) 70% alcohol
    d) Betadine – iodine
20. The following are effects of shortwave diathermy machine. Which one is not?
    a) Increase heart beat rate
    b) Increase blood circulation
    c) Increase metabolic rate
    d) Protein synthesis
21. The common short wave diathermy application techniques are?
    a) Capacitor and reflective method
    b) Condenser and reflective method
    c) Capacitor plate and inductive method
    d) Inductive and reflective method

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22. What is the application of diathermy?
    a) Surgical and therapeutic
    b) Therapeutic and diagnostic
    c) Diagnostic and surgical
    d) Diagnostic and rehabilitation
23. Which one of the following diathermy machine is good for deep tissue healing?
    a) Short wave
    b) Ultrasound
    c) Cold compress
    d) Microwave
24. The process in which a photon is absorbed by the atom, causing an electron to jump from a lower energy level to a higher energy level is referred to as?
    a) Absorption of Radiation
    b) Spontaneous Emission
    c) Stimulated Emission
    d) Population inversion
25. Which endoscope can be used to examine the knees before and after a surgery?
    a) Colonoscopy
    b) Arthroscopy
    c) Bronchoscopy
    d) Laryngoscopy
26. Which one of the following is not a structural requirement of a dental chair?
    a) Should be strong and robust
    b) Noiseless and easy to operate
    c) Relatively cheap and durable
    d) Should be able to hold up to 80kg of weight
27. Short wave diathermy produces microwaves from;
    a) Ultraviolet radiation
    b) Thermal radiation
    c) Electromagnetic radiation
    d) Electrical radiation
28. Which of the below equipment is not found in the labour room?
    a) Suction machine
    b) Baby warmer
    c) Delivery bed
    d) Weighing scale

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29. Which one of the following statement is not correct about features of electric delivery bed?
    a) Back up battery
    b) Easy to operate head down tilt
    c) Wide choice of safe and comfortable position that can be adjusted during delivery
    d) Limited movement
30. The following sunction devices are commonly used in dental unit, which one is not?
    a) Internal suction pump
    b) External suction pump
    c) Jet pump
    d) Central suction unit
31. What is the pressure range of compressed air supply to dental chair?
    a) 5.2 to 7.5 bars
    b) 8.5 to 10 bars
    c) 7.5 to 9.5 bars
    d) 2 to 5.5 bars
32. How do jet nebulizers convert liquid medication to droplets?
    a) By spraying a jet of water through the drug
    b) By evaporating the liquid in to vapour
    c) By ultrasound waves
    d) By pushing jets of compressed air through the liquid drug to aerosolize the medication
33. Which light is most effective for phototherapy
    a) Blue white light
    b) Blue light
    c) Blue green light
    d) Green white light
34. Dental chair should have at least the following movements, except?
    a) Tandenburg
    b) Upward movement
    c) Base rotation movement
    d) Backrest rotation movement
35. What causes tooth decay
    a) Acid
    b) Age
    c) Cavities
    d) Caffeine

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36. Which of the following is not a dental unit accessories?
   a) Ultrasonic scaler
   b) Surgery hand piece
   c) Fibre optic light
   d) Sprayvit
37. Which of the following is not a hydraulic component in a Hydro-mechanical chair?
   a) Limit switches
   b) Directional valves
   c) Non-return valves
   d) Limiting valves
38. Which one is not a type of galvanic waveform?
   a) Rectangular surging
   b) Trapezoidal surging
   c) Triangular surging
   d) Square surging
39. Which one of the following is not a type of electrode used in short wave diathermy?
   a) Flexible pads
   b) Space plates
   c) The bimode plate
   d) Coil or cable
40. Which among the following dental chairs is the odd out?
   a) Mechanical chair
   b) Hydro-electrical chair
   c) Hydro-mechanical chair
   d) Electro-mechanical chair

SECTION 2: SHORT ANSWER QUESTIONS (40 MARKS)

41. (a) State the cause of electrocautery induced electrical interference on the electrocardiogram machine. (1mk)
   (b) State two types of electrosurgical technique. (2mks)
   (c) Name two main operational systems of basic mechanical ventilator. (2mks)
42. Name five parts of a manually operated vacuum extractor. (5mks)
43. (a) State the application of laser in ophthalmology. (2mks)
   (b) Name three subsystems that make up an endoscope. (3mks)
44. State five factors to be considered in designing of a dental office. (5mks)
45. State five physiotherapy equipment. (5mks)
46. State five demerits of Electro-Mechanical dental chair. (5mks)
47. Highlight five structural requirements of a Dental Chair. (5mks)
48. (a) State two advantages of an indirect ophthalmoscope over a direct ophthalmoscope. (2mks)

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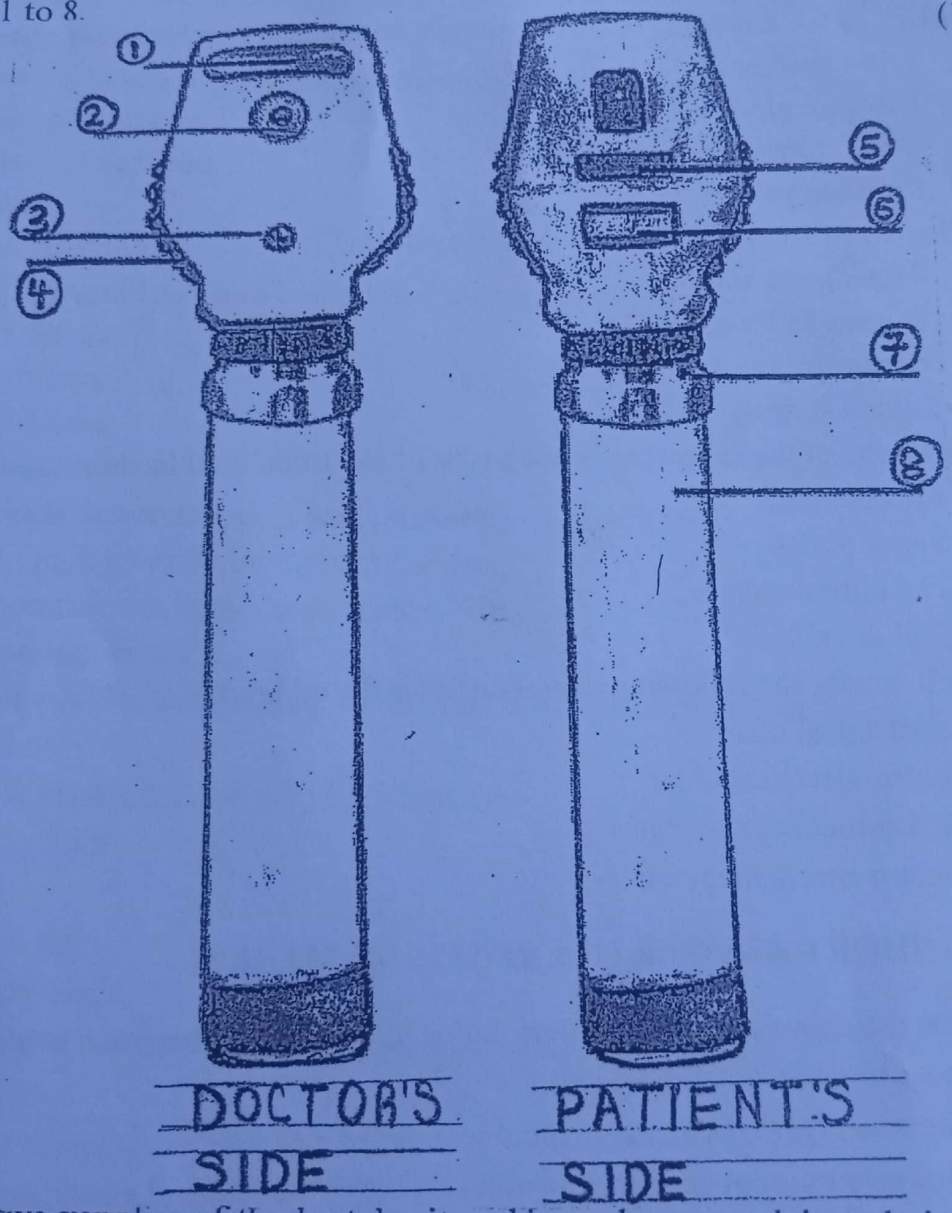
(b) Highlight the three major subsystems of a modern slit lamp. (3mks)

SECTION 3: LONG ANSWER QUESTION (20 MARKS)

49. (a) Figure 1 below is a direct ophthalmoscope. Explain its functions and name the parts labelled 1 to 8. (10 marks)



(b) State the five supplies of the dental unit and in each case explain technical requirements and purpose. (10 marks)